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L04: Text Representation

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9/19/2026

Lab 04: Text Representation

Throughout this lab, I learned how text can be converted into numerical representations that machine learning models can understand. I explored Bag of Words (BOW), TF-IDF, N-grams, and word embeddings and learned the advantages and limitations of each method.

BOW represents documents by counting words, but it does not understand word order or context and can create sparse data. TF-IDF improves on this by giving more importance to words that are useful for distinguishing between documents. N-grams can capture word combinations such as “not good,” providing more context than individual words.

The most interesting part was word embeddings. They represent words as numerical vectors and can capture relationships between words. The example “king - man + woman $\approx$ queen” showed how word relationships can be represented mathematically.

I also used these text representations for movie review classification. **[Insert your best-performing method here]** performed best in my results. This showed me that the best representation depends on the type of data and task.

Overall, this lab helped me understand how different text representation methods are used in real-world applications such as sentiment analysis, search engines, spam detection, and chatbots. I also learned that biased training data can lead to biased results. If I continue studying NLP, I would like to learn more about contextual embeddings and transformer models.